

Dielectric–dielectric boundary conditions in electrostatics

Consider a planar interface between two linear dielectrics:

Medium 1: ε_1 (above), Medium 2: ε_2 (below).

Let $\hat{\mathbf{n}}$ be the unit normal pointing from medium 2 to medium 1. We decompose fields into normal and tangential components:

$$\mathbf{E} = \mathbf{E}_t + E_n \hat{\mathbf{n}}, \quad \mathbf{D} = \mathbf{D}_t + D_n \hat{\mathbf{n}}.$$

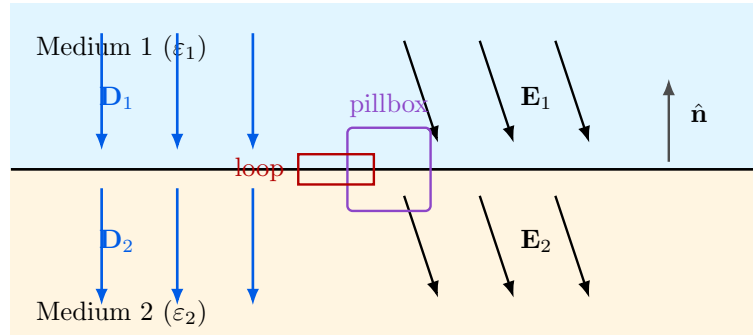


Figure 1: Planar dielectric boundary: pillbox gives the normal- $\mathbf{D}$ condition; small loop gives tangential- $\mathbf{E}$ condition (electrostatics).

Boundary condition 1: normal component (use Gauss's law)

If we apply Gauss's law directly to $\mathbf{E}$.

$$\nabla \cdot \mathbf{E} = \frac{\rho_f + \rho_b}{\varepsilon_0}$$

so at a dielectric boundary, $\mathbf{E}$ can show a jump because bound charges may exist at the interface. That is why it is not convenient to state the normal boundary condition using $\mathbf{E}$ alone.

Apply Gauss's law to $\mathbf{D}$ By definition, $\nabla \cdot \mathbf{D} = \rho_f$. Integrate over the pillbox that straddles the boundary:

$$\iiint_V (\nabla \cdot \mathbf{D}) dV = \oiint_S \mathbf{D} \cdot d\mathbf{S} = \iint_{\text{top}} \mathbf{D}_1 \cdot \hat{\mathbf{n}} dS + \iint_{\text{bottom}} \mathbf{D}_2 \cdot (-\hat{\mathbf{n}}) dS = Q_{f,\text{enc}}.$$

Let the pillbox area be A and its height $\rightarrow 0$, so side flux vanishes. Then

$$(D_{1n} - D_{2n})A = \sigma_f A \quad \Rightarrow \quad \boxed{D_{1n} - D_{2n} = \sigma_f} \quad (\text{BC1})$$

where σ_f is the **free surface charge density** at the boundary.



Special case : no free charge at the boundary. If $\sigma_f = 0$,

$$\boxed{D_{1n} = D_{2n}} \quad (\text{normal component of } \mathbf{D} \text{ is continuous}).$$

If a free surface charge exists at the boundary. If $\sigma_f \neq 0$,

$$\boxed{D_{1n} - D_{2n} = \sigma_f} \quad (\text{discontinuity equals the free surface charge}).$$

Relating to $\mathbf{E}$ (linear isotropic media). If each medium is linear and isotropic, $\mathbf{D} = \varepsilon \mathbf{E}$, hence

$$\varepsilon_1 E_{1n} - \varepsilon_2 E_{2n} = \sigma_f.$$

In particular, if $\sigma_f = 0$, then

$$\boxed{\varepsilon_1 E_{1n} = \varepsilon_2 E_{2n}} \Rightarrow \text{normal component of } \mathbf{E} \text{ generally changes across the interface.}$$

Boundary condition 2: tangential component (use $\oint \mathbf{E} \cdot d\mathbf{l} = 0$)

In electrostatics,

$$\boxed{\nabla \times \mathbf{E} = \mathbf{0}} \quad \Longleftrightarrow \quad \boxed{\oint \mathbf{E} \cdot d\mathbf{l} = 0}$$

for any closed loop.

Take a very small rectangular loop that straddles the boundary, with its long sides parallel to the interface. Let the tangential direction be $\hat{\mathbf{t}}$ and the loop length along the interface be Δl . As the loop height $\rightarrow 0$, contributions from the short sides vanish, hence

$$(\mathbf{E}_{1t} \cdot \hat{\mathbf{t}}) \Delta l - (\mathbf{E}_{2t} \cdot \hat{\mathbf{t}}) \Delta l = 0 \Rightarrow \boxed{E_{1t} = E_{2t}} \quad (\text{BC2})$$

So:

Tangential component of $\mathbf{E}$ is continuous at a dielectric boundary (electrostatics).

Summary (boundary conditions in electrostatics)

- **Normal component (from Gauss law in $\mathbf{D}$):**

$$\boxed{D_{1n} - D_{2n} = \sigma_f} \Rightarrow \text{if } \sigma_f = 0, \boxed{D_{1n} = D_{2n}}.$$

- **Tangential component (from $\oint \mathbf{E} \cdot d\mathbf{l} = 0$):**

$$\boxed{E_{1t} = E_{2t}} \quad (\text{always true in electrostatics}).$$

- For linear isotropic media: $\mathbf{D} = \varepsilon \mathbf{E}$, so if $\sigma_f = 0$ then

$$\boxed{\varepsilon_1 E_{1n} = \varepsilon_2 E_{2n}} \quad (\text{normal-}\mathbf{E} \text{ generally changes}).$$

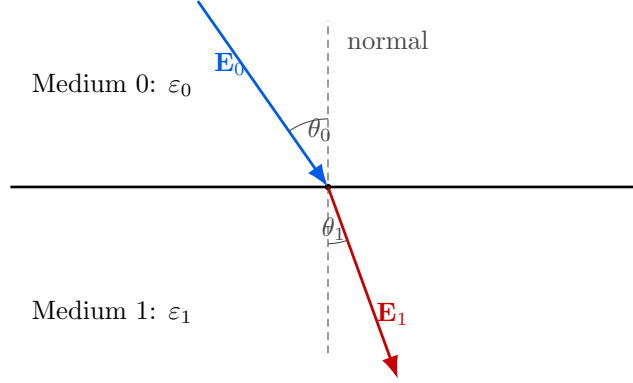


Figure 2: Field-line (ray) picture at a dielectric boundary: $\mathbf{E}_0$ approaches the interface at incidence angle θ_0 , then bends and leaves as $\mathbf{E}_1$ at refraction angle θ_1 (angles measured w.r.t. the normal).

Step 1: Write the two boundary conditions

Since $\sigma_f = 0$,

$$\boxed{D_{0n} = D_{1n}} \Rightarrow \boxed{\varepsilon_0 E_0 \cos \theta_0 = \varepsilon_1 E_1 \cos \theta_1} \quad (1)$$

Also, tangential $\mathbf{E}$ is continuous in electrostatics:

$$\boxed{E_{0t} = E_{1t}} \Rightarrow \boxed{E_0 \sin \theta_0 = E_1 \sin \theta_1} \quad (2)$$

Step 2: “Snell-like” relation for θ_1

Divide (2) by (1):

$$\frac{E_0 \sin \theta_0}{\varepsilon_0 E_0 \cos \theta_0} = \frac{E_1 \sin \theta_1}{\varepsilon_1 E_1 \cos \theta_1} \Rightarrow \boxed{\frac{1}{\varepsilon_0} \tan \theta_0 = \frac{1}{\varepsilon_1} \tan \theta_1} \quad (3)$$

Hence

$$\boxed{\tan \theta_1 = \frac{\varepsilon_1}{\varepsilon_0} \tan \theta_0}$$

(So θ_1 can be computed once $\theta_0, \varepsilon_0, \varepsilon_1$ are known.)

Step 3: Find E_1

From (1),

$$\varepsilon_0 E_0 \cos \theta_0 = \varepsilon_1 E_1 \cos \theta_1 \Rightarrow \boxed{E_1 \cos \theta_1 = \frac{\varepsilon_0}{\varepsilon_1} E_0 \cos \theta_0} \quad (4)$$

From (2),

$$\boxed{E_1 \sin \theta_1 = E_0 \sin \theta_0} \quad (5)$$

Now square and add (4) and (5):

$$E_1^2(\cos^2 \theta_1 + \sin^2 \theta_1) = \left(\frac{\varepsilon_0}{\varepsilon_1} E_0 \cos \theta_0 \right)^2 + (E_0 \sin \theta_0)^2$$

$$\Rightarrow \boxed{E_1 = E_0 \sqrt{\left(\frac{\varepsilon_0}{\varepsilon_1} \right)^2 \cos^2 \theta_0 + \sin^2 \theta_0}} \quad (6)$$

This is exactly the final expression in the attachment.

Special case (as written): normal incidence

If $\theta_0 = 0$,

$$E_1 = E_0 \left(\frac{\varepsilon_0}{\varepsilon_1} \right).$$

So at normal incidence the field magnitude is reduced by the factor $\varepsilon_1/\varepsilon_0$ when entering the higher-permittivity medium.

Takeaway

- Tangential component of $\mathbf{E}$ is continuous: $E_0 \sin \theta_0 = E_1 \sin \theta_1$ (no reduction).
- Normal component is scaled by permittivity: $\varepsilon_0 E_0 \cos \theta_0 = \varepsilon_1 E_1 \cos \theta_1$.
- Hence,

$$E_1 = E_0 \sqrt{\left(\frac{\varepsilon_0}{\varepsilon_1} \right)^2 \cos^2 \theta_0 + \sin^2 \theta_0}, \quad \tan \theta_1 = \frac{\varepsilon_1}{\varepsilon_0} \tan \theta_0.$$

- **Important special case: normal incidence** ($\theta_0 = 0$). If $\theta_0 = 0$, then $E_{0t} = 0$ and the entire field is normal. From $\varepsilon_0 E_0 \cos \theta_0 = \varepsilon_1 E_1 \cos \theta_1$ with $\cos \theta_0 = \cos \theta_1 = 1$,

$$\boxed{E_1 = E_0 \frac{\varepsilon_0}{\varepsilon_1}}$$

So the field magnitude in medium 1 is reduced by the **dielectric constant ratio**.

- **Matching-angle remark** From the expression

$$E_1 = E_0 \sqrt{\left(\frac{\varepsilon_0}{\varepsilon_1} \right)^2 \cos^2 \theta_0 + \sin^2 \theta_0},$$

as θ_0 increases, the $\sin^2 \theta_0$ term becomes more dominant. Hence there can exist a value of θ_0 for which

$$\boxed{E_1 = E_0}$$

(i.e., a “matching” condition where the magnitude does not change across the boundary).

Some more problems on capacitance

Problem 1: Parallel-plate capacitor with a dielectric slab (air–dielectric–air)

Given Two parallel metal plates of area A are separated by total distance d . The top plate is at potential V_0 and the bottom plate is grounded. The gap is filled as:

Region (1): air of thickness z_0 , Region (2): dielectric of thickness $(z_1 - z_0)$,

Region (3): air of thickness $(d - z_1)$,

where $0 < z_0 < z_1 < d$. The dielectric has relative permittivity ϵ_r so $\epsilon = \epsilon_r \epsilon_0$.

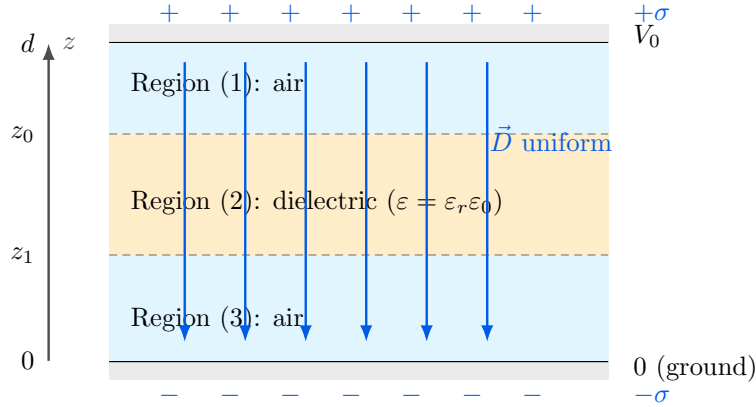


Figure 3: Layered capacitor: air–dielectric–air. $\vec{D}$ is uniform (set by free charge), but $\vec{E}$ differs in each region.

Key idea: use $\nabla \cdot \vec{D} = \rho_f$

Because polarization creates induced charges inside the dielectric, it is cleaner to use the displacement vector:

$$\nabla \cdot \vec{D} = \rho_f \quad \Rightarrow \quad \iint \vec{D} \cdot d\vec{A} = Q_{\text{free,enc}}.$$

Take a Gaussian pillbox of area A that cuts the top metal plate. Only the face in the gap contributes, so

$$D_n A = \sigma A \quad \Rightarrow \quad \boxed{D_n = \sigma}.$$

Since there is **no free charge at the internal air–dielectric interfaces**, the normal component of $\vec{D}$ is continuous there as well. Hence

$$\boxed{D_{1z} = D_{2z} = D_{3z} = \sigma} \quad \Rightarrow \quad \vec{D} \text{ is uniform across Regions (1),(2),(3).}$$

Electric field in each region

Use $\vec{D} = \epsilon \vec{E}$ in each medium.



Region (1): air (permittivity ε_0).

$$D_{1z} = \varepsilon_0 E_{1z} = \sigma \quad \Rightarrow \quad \boxed{E_{1z} = \frac{\sigma}{\varepsilon_0}}$$

Region (2): dielectric (permittivity $\varepsilon = \varepsilon_r \varepsilon_0$).

$$D_{2z} = \varepsilon E_{2z} = \sigma \quad \Rightarrow \quad \boxed{E_{2z} = \frac{\sigma}{\varepsilon_r \varepsilon_0}}$$

Region (3): air (permittivity ε_0).

$$D_{3z} = \varepsilon_0 E_{3z} = \sigma \quad \Rightarrow \quad \boxed{E_{3z} = \frac{\sigma}{\varepsilon_0}}$$

So: $\vec{D}$ is uniform, but $\vec{E}$ is *not* uniform because ε changes.

Potential difference V_0 (sum of drops across each region)

The total drop is the integral across the full separation:

$$V_0 = \int_0^d E_z dz = \int_0^{z_1} E_z dz + \int_{z_1}^d E_z dz = \int_0^{z_1} \frac{\sigma}{\varepsilon_0} dz + \int_{z_1}^{z_0} \frac{\sigma}{\varepsilon_r \varepsilon_0} dz + \int_{z_0}^d \frac{\sigma}{\varepsilon_0} dz.$$

Therefore,

$$\boxed{V_0 = \frac{\sigma}{\varepsilon_0} z_0 + \frac{\sigma}{\varepsilon_r \varepsilon_0} (z_1 - z_0) + \frac{\sigma}{\varepsilon_0} (d - z_1)}$$

or equivalently,

$$\boxed{V_0 = \frac{\sigma}{\varepsilon_0} \left[(d - (z_1 - z_0)) + \frac{(z_1 - z_0)}{\varepsilon_r} \right].}$$

Capacitance

Charge on the plates: $Q = \sigma A$. Hence

$$C = \frac{Q}{V_0} = \frac{\sigma A}{V_0}.$$

Using the expression above,

$$\boxed{C = \frac{A}{\frac{z_0}{\varepsilon_0} + \frac{(z_1 - z_0)}{\varepsilon_r \varepsilon_0} + \frac{(d - z_1)}{\varepsilon_0}} = \frac{\varepsilon_0 A}{z_0 + \frac{(z_1 - z_0)}{\varepsilon_r} + (d - z_1)}}$$

Special case (a): entire gap filled with dielectric

If the dielectric fills the whole gap: $z_0 = 0$ and $z_1 = d$, then

$$C = \frac{\varepsilon_0 A}{d/\varepsilon_r} = \boxed{\frac{\varepsilon_r \varepsilon_0 A}{d}}$$

So capacitance increases from $\varepsilon_0 A/d$ to $\varepsilon_r \varepsilon_0 A/d$.

Takeaway

- In layered dielectrics with no free charge at internal interfaces: $D_n = \sigma$ is uniform.
- Electric field changes by region: $E = \sigma/\varepsilon$. Hence E is smaller inside the dielectric.
- The total voltage is the **sum of drops** across each thickness:

$$V_0 = \sum_i E_i (\text{thickness})_i.$$

- Final result (air–dielectric–air):

$$C = \frac{\varepsilon_0 A}{z_0 + \frac{(z_1 - z_0)}{\varepsilon_r} + (d - z_1)}$$

So C depends only on geometry and materials (layer thicknesses + permittivities).

What is happening here when a dielectric (ε_r large) fills most of the gap?

Setup. A parallel-plate capacitor (area A) is connected to a source that keeps the top plate at V_0 and the bottom plate at 0. Along the field direction (take x from bottom plate to top plate), the gap is:

$$\text{air: } 0 < x < x_0, \quad \text{dielectric: } x_0 < x < x_1, \quad \text{air: } x_1 < x < d,$$

where d is total plate separation and the dielectric has $\varepsilon = \varepsilon_r \varepsilon_0$.

Step 1: Why we talk about an “average” field

Because the source fixes the voltage,

$$\boxed{V_0 = \int_0^d E(x) dx} \Rightarrow \boxed{E_{\text{avg}} = \frac{1}{d} \int_0^d E(x) dx = \frac{V_0}{d}}$$

So *whatever* the field distribution is, its integral must equal V_0 .



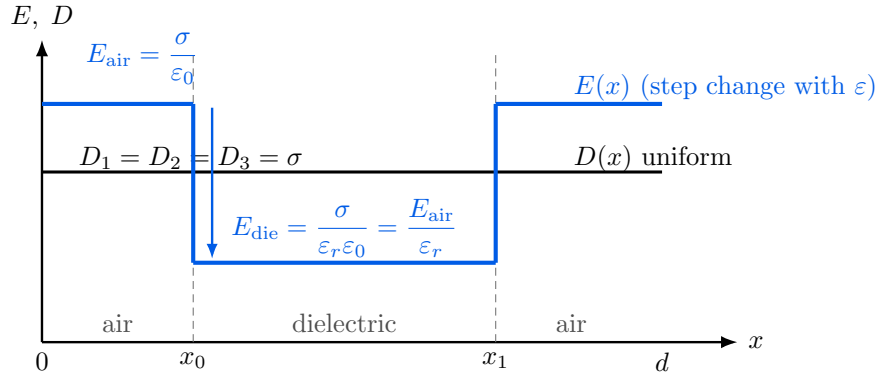


Figure 4: Clear labels: $D(x)$ is uniform at σ , while $E(x)$ shows a step reduction inside dielectric and returns to the air value in the air gaps.

Step 2: Why E becomes small inside dielectric and large in the air gaps

Use displacement vector:

$$\nabla \cdot \vec{D} = \rho_f.$$

There is free charge only on the metal plates (no free charge at the air–dielectric interfaces), so the normal component of $\vec{D}$ is the same in all three regions:

$$D_{air} = D_{die} = \sigma$$

where σ is the free surface charge density on the plates.

Now $\vec{D} = \epsilon \vec{E}$ in each region:

$$E_{air} = \frac{D}{\epsilon_0} = \frac{\sigma}{\epsilon_0}, \quad E_{die} = \frac{D}{\epsilon_r \epsilon_0} = \frac{\sigma}{\epsilon_r \epsilon_0} = \frac{E_{air}}{\epsilon_r}.$$

Hence E is much smaller in dielectric (by ϵ_r), and much larger in air.

Step 3: Voltage constraint forces the “compensation”

Compute $V_0 = \int_0^d E dx$ using piecewise-constant fields:

$$V_0 = E_{air} (x_0 - 0) + E_{die} (x_1 - x_0) + E_{air} (d - x_1).$$

Substitute $E_{die} = E_{air}/\epsilon_r$:

$$V_0 = E_{air} \left[(x_0) + (d - x_1) + \frac{(x_1 - x_0)}{\epsilon_r} \right].$$

Therefore

$$E_{air} = \frac{V_0}{x_0 + (d - x_1) + \frac{(x_1 - x_0)}{\epsilon_r}} \quad E_{die} = \frac{E_{air}}{\epsilon_r}$$



Now note the key inequality (this is exactly the “compensation” statement):

$$x_0 + (d - x_1) + \frac{(x_1 - x_0)}{\varepsilon_r} < x_0 + (d - x_1) + (x_1 - x_0) = d \quad (\varepsilon_r > 1)$$

so

$$E_{air} > \frac{V_0}{d} = E_{avg}, \quad E_{die} < E_{avg}.$$

Meaning: the field inside the dielectric is below average, and the air-gap fields become above average so that the integral still gives V_0 .

Step 4: Why this implies “more free charge” and hence higher capacitance

From above,

$$\sigma = \varepsilon_0 E_{air} = \frac{\varepsilon_0 V_0}{x_0 + (d - x_1) + \frac{(x_1 - x_0)}{\varepsilon_r}}.$$

Thus charge on plates $Q = \sigma A$ becomes

$$Q = \frac{\varepsilon_0 A V_0}{x_0 + (d - x_1) + \frac{(x_1 - x_0)}{\varepsilon_r}}.$$

So capacitance $C = Q/V_0$ is

$$C = \frac{\varepsilon_0 A}{x_0 + (d - x_1) + \frac{(x_1 - x_0)}{\varepsilon_r}}$$

As ε_r increases and/or the dielectric occupies more thickness $(x_1 - x_0)$, the denominator decreases, so C increases. This is exactly means the “more induced/free charge $\Rightarrow$ increased capacitance” (battery supplies extra free charge to maintain V_0).

Takeaway

- Battery fixes $V_0 \Rightarrow V_0 = \int_0^d E(x) dx$ and $E_{avg} = V_0/d$.
- No free charge at internal interfaces $\Rightarrow D_n$ is the same in all layers: $D = \sigma$.
- Because $E = D/\varepsilon$: $E_{die} = E_{air}/\varepsilon_r$ (field drops inside dielectric).
- To still satisfy $\int E dx = V_0$, the air-gap field becomes larger than average:

$$E_{air} > E_{avg}, \quad E_{die} < E_{avg}.$$

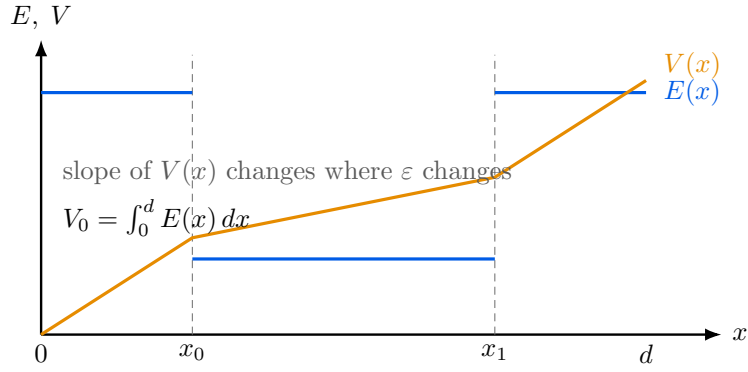


Figure 5: Qualitative plot: $E(x)$ is high in air gaps and low in dielectric; $V(x)$ is continuous but piecewise linear.

- With battery connected, $\sigma = \epsilon_0 E_{air}$ increases $\Rightarrow Q = \sigma A$ increases $\Rightarrow$

$$C = \frac{Q}{V_0} = \frac{\epsilon_0 A}{x_0 + (d - x_1) + \frac{(x_1 - x_0)}{\epsilon_r}} \text{ increases.}$$

Problem 2: Multi-layer capacitor (air- ϵ_1 -air- ϵ_2 -air- ϵ_3)

Given Two large parallel metal plates of area A are separated along x (bottom plate at $x = 0$, top plate at $x = b$). The top plate is at potential V_0 and the bottom plate is grounded. Between the plates we have alternating layers:

$$\underbrace{\epsilon_0}_{x_0 \rightarrow x_1}, \underbrace{\epsilon_1}_{x_1 \rightarrow x_2}, \underbrace{\epsilon_0}_{x_2 \rightarrow x_3}, \underbrace{\epsilon_2}_{x_3 \rightarrow x_4}, \underbrace{\epsilon_0}_{x_4 \rightarrow x_5}, \underbrace{\epsilon_3}_{x_5 \rightarrow x_6}, \quad 0 = x_0 < x_1 < \dots < x_6 = b.$$

(Here ϵ_0 denotes air/vacuum layers; $\epsilon_1, \epsilon_2, \epsilon_3$ are the slab permittivities.)

Step 1: Use displacement vector (same method as Problem 1)

Because induced charges exist inside dielectric slabs, we use

$$\nabla \cdot \vec{D} = \rho_f.$$

Free charge exists only on the metal plates, so the normal component of $\vec{D}$ is the same in every layer:

$$D_1 = D_2 = \dots = D_6 = \sigma$$

where σ is the free surface charge density on the plates. Hence in layer i ,

$$\vec{E}_i = \frac{\vec{D}}{\epsilon_i} \Rightarrow E_i = \frac{\sigma}{\epsilon_i}$$



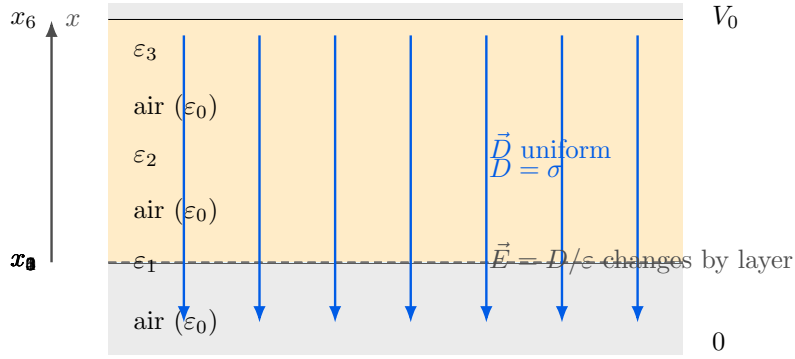


Figure 6: Multi-layer stack: $\vec{D}$ is uniform (set by free charge), while $\vec{E}$ is piecewise constant and changes whenever ε changes.

with $\varepsilon_i \in \{\varepsilon_0, \varepsilon_1, \varepsilon_0, \varepsilon_2, \varepsilon_0, \varepsilon_3\}$.

Step 2: Add voltage drops across each layer

Total potential difference is the integral of $E(x)$:

$$V_0 = \int_0^6 E(x) dx = \sum_{k=1}^6 E_k (x_k - x_{k-1}).$$

Substitute $E_k = \sigma/\varepsilon_k$:

$$V_0 = \frac{\sigma}{\varepsilon_0}(x_1 - x_0) + \frac{\sigma}{\varepsilon_1}(x_2 - x_1) + \frac{\sigma}{\varepsilon_0}(x_3 - x_2) + \frac{\sigma}{\varepsilon_2}(x_4 - x_3) + \frac{\sigma}{\varepsilon_0}(x_5 - x_4) + \frac{\sigma}{\varepsilon_3}(x_6 - x_5)$$

Step 3: Capacitance

Charge on the plates:

$$Q = \sigma A.$$

Therefore

$$C = \frac{Q}{V_0} = \frac{\sigma A}{V_0}.$$

Using the V_0 expression above:

$$C = \frac{A}{\frac{(x_1-x_0)}{\varepsilon_0} + \frac{(x_2-x_1)}{\varepsilon_1} + \frac{(x_3-x_2)}{\varepsilon_0} + \frac{(x_4-x_3)}{\varepsilon_2} + \frac{(x_5-x_4)}{\varepsilon_0} + \frac{(x_6-x_5)}{\varepsilon_3}}$$

This is the standard “series combination” result:

$$\frac{1}{C} = \sum_{k=1}^6 \frac{(x_k - x_{k-1})}{\varepsilon_k A}$$



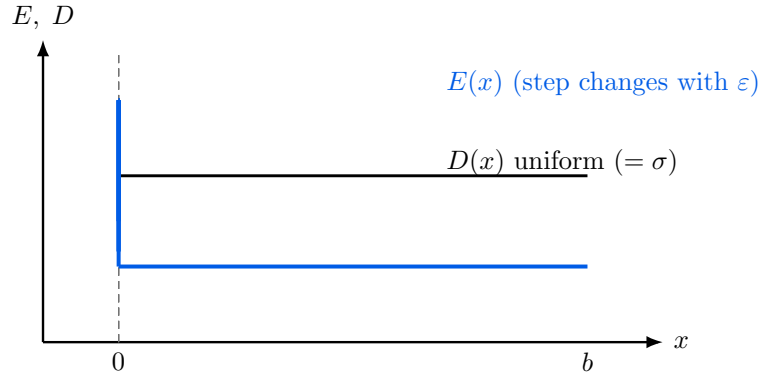


Figure 7: Step-plot view: D constant; $E = \sigma/\varepsilon$ changes in each layer.

Takeaway (Problem 3)

- Use $\nabla \cdot \vec{D} = \rho_f \Rightarrow$ in all layers (no free charge at internal boundaries), $D = \sigma$ is uniform.
- In each layer i : $E_i = \sigma/\varepsilon_i$. So larger $\varepsilon_i \Rightarrow$ smaller E_i .
- Voltage adds in series:

$$V_0 = \sum_i E_i t_i = \sigma \sum_i \frac{t_i}{\varepsilon_i}$$

where t_i is the thickness of layer i .

- Final capacitance:

$$C = \frac{Q}{V_0} = \frac{A}{\sum_i \frac{t_i}{\varepsilon_i}} \quad \left(\text{equivalently } \frac{1}{C} = \sum_i \frac{t_i}{\varepsilon_i A} \right)$$

Energy stored in electrostatics (not discussed in class)

We briefly add this topic for completeness.

1) Starting point: stored energy written using charge and potential

The total electrostatic energy can be written as

$$W = \frac{1}{2} \int_V \rho \phi dv.$$

In free space (no dielectric), one can also show

$$W = \frac{1}{2} \int_V \epsilon_0 |\vec{E}|^2 dv,$$

and this is the familiar “energy in the field” form.

2) With dielectric present: we use $\nabla \cdot \vec{D} = \rho_f$

When dielectric is present, it is safer to separate *free* charge and *bound* charge. The governing Gauss-law form is

$$\nabla \cdot \vec{D} = \rho_f,$$

where ρ_f is the free-charge density.

Therefore,

$$W = \frac{1}{2} \int_V \rho_f \phi dv = \frac{1}{2} \int_V (\nabla \cdot \vec{D}) \phi dv.$$

3) Key identity used in the note

Use the product-rule identity (exactly the one implied in your derivation):

$$\nabla \cdot (\phi \vec{D}) = \phi (\nabla \cdot \vec{D}) + \vec{D} \cdot (\nabla \phi).$$

Rearrange:

$$\phi (\nabla \cdot \vec{D}) = \nabla \cdot (\phi \vec{D}) - \vec{D} \cdot (\nabla \phi).$$

But $\vec{E} = -\nabla \phi$, so $-\vec{D} \cdot \nabla \phi = \vec{D} \cdot \vec{E}$. Hence

$$\phi (\nabla \cdot \vec{D}) = \nabla \cdot (\phi \vec{D}) + \vec{D} \cdot \vec{E}.$$

4) Convert volume integral using divergence theorem

Substitute into the energy expression:

$$W = \frac{1}{2} \int_V \left[\nabla \cdot (\phi \vec{D}) + \vec{D} \cdot \vec{E} \right] dv.$$

Using the divergence theorem,

$$\int_V \nabla \cdot (\phi \vec{D}) dv = \oint_S \phi \vec{D} \cdot d\vec{S}.$$

Therefore,

$$W = \frac{1}{2} \oint_S \phi \vec{D} \cdot d\vec{S} + \frac{1}{2} \int_V \vec{D} \cdot \vec{E} dv.$$

- The $\oint_S \phi \vec{D} \cdot d\vec{S}$ term is a **surface term** (depends on how the device is bounded / what conductor surfaces we include).
- The $\frac{1}{2} \int_V \vec{D} \cdot \vec{E} dv$ term represents **energy stored in the field**.

Energy density (main result)

$$u = \frac{1}{2} \vec{D} \cdot \vec{E}$$

So for linear isotropic dielectric $\vec{D} = \epsilon \vec{E}$,

$$u = \frac{1}{2} \epsilon |\vec{E}|^2 \quad (\text{instead of } \frac{1}{2} \epsilon_0 |\vec{E}|^2).$$

5) Important note

For linear dielectric, $\vec{D} = \epsilon \vec{E} = \epsilon_r \epsilon_0 \vec{E}$. Then

$$u = \frac{1}{2} \vec{D} \cdot \vec{E} = \frac{1}{2} \epsilon |\vec{E}|^2 = \frac{1}{2} \epsilon_r \epsilon_0 |\vec{E}|^2 = \frac{|\vec{D}|^2}{2\epsilon}.$$

Hence:

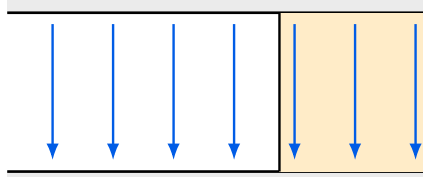
- If $\vec{E}$ is held constant (e.g., voltage fixed, same geometry), then

$$u \propto \epsilon \Rightarrow u \text{ increases with } \epsilon_r.$$

- If $\vec{D}$ is held constant (e.g., free charge fixed), then

$$u = \frac{D^2}{2\epsilon} \Rightarrow u \text{ decreases with } \epsilon_r.$$

(i) Voltage fixed (V_0 constant) $\Rightarrow \vec{E}$ roughly fixed



(ii) Charge fixed (Q constant) $\Rightarrow \vec{D}$ fixed

Figure 8: Inserting a dielectric slab: energy change depends on what is held constant (voltage or charge).

Takeaway (energy vs. dielectric)

- Energy density in a dielectric: $u = \frac{1}{2} \vec{D} \cdot \vec{E}$.
- Linear dielectric: $u = \frac{1}{2} \epsilon E^2 = \frac{D^2}{2\epsilon}$.
- V fixed $\Rightarrow E$ fixed: inserting dielectric $\Rightarrow u \uparrow$ by factor ϵ_r .
- Q fixed $\Rightarrow D$ fixed: inserting dielectric $\Rightarrow u \downarrow$ by factor ϵ_r .

Only for extra reading

What exactly is the surface term, and why does it depend on the device boundary?

From the derivation we obtained

$$W = \frac{1}{2} \oint_S \phi \vec{D} \cdot d\vec{S} + \frac{1}{2} \int_V \vec{D} \cdot \vec{E} dv.$$

The second term is naturally interpreted as *energy in the field inside the volume V*. The first term is a *surface term* because it only needs information on the boundary S .

Why does $\oint_S \phi \vec{D} \cdot d\vec{S}$ depend on what surfaces we include? Because $\vec{D} \cdot d\vec{S}$ counts *free charge* that pierces the chosen boundary. Indeed, Gauss's law for $\vec{D}$ says

$$\oint_S \vec{D} \cdot d\vec{S} = Q_f^{(\text{enclosed})}.$$

So on each conductor surface that lies on S , we have

$$\vec{D} \cdot \hat{n} = \sigma_f, \quad \Rightarrow \quad \int_{S_k} \vec{D} \cdot d\vec{S} = \int_{S_k} \sigma_f dA = Q_{f,k}.$$

If that conductor is an electrode at a (nearly) constant potential ϕ_k , then on that electrode surface,

$$\int_{S_k} \phi \vec{D} \cdot d\vec{S} = \phi_k \int_{S_k} \vec{D} \cdot d\vec{S} = \phi_k Q_{f,k}.$$

Therefore the surface term becomes, very explicitly,

$$\oint_S \phi \vec{D} \cdot d\vec{S} = \sum_{\text{electrodes } k \subset S} \phi_k Q_{f,k} + \int_{S_\infty} \phi \vec{D} \cdot d\vec{S}.$$

Two common cases (this is the key logical point):



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1. **If S is chosen to be a very large surface at infinity** and we use the reference $\phi(\infty) = 0$, then typically $\phi \rightarrow 0$ fast enough and $\vec{D}$ decays so that

$$\int_{S_\infty} \phi \vec{D} \cdot d\vec{S} \approx 0.$$

Then the surface term reduces to only the electrode contributions:

$$\oint_S \phi \vec{D} \cdot d\vec{S} = \sum_k \phi_k Q_{f,k}.$$

This is essentially the familiar “work to assemble charge on conductors at given potentials”.

2. **If S is chosen to cut through electrodes or does not include all relevant conductors**, then the sum changes, because you have changed which free charges $Q_{f,k}$ are counted as “enclosed” and what value of ϕ multiplies them. That is why it is called a *surface term*: it depends on how you close the volume around the device.

Practical takeaway. The volume term $\frac{1}{2} \int_V \vec{D} \cdot \vec{E} dv$ is the *field energy inside the chosen volume*. If you choose V to include only part of space (e.g. only inside the dielectric slab region), then energy stored in fields outside that region is not counted in the volume term, and the surface term compensates for that bookkeeping. If you choose V to include all space (bounded by S_∞ with $\phi(\infty) = 0$), the surface-at-infinity contribution vanishes and the only remaining surface contributions come from electrodes: $\sum_k \phi_k Q_{f,k}$.